

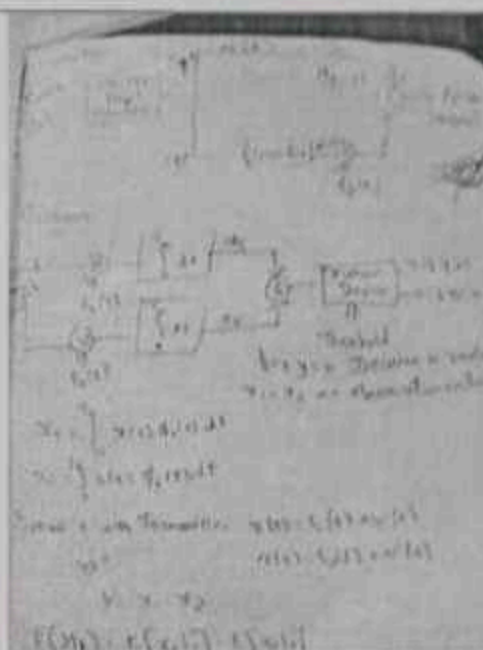
RAMCO INSTITUTE OF TECHNOLOGY
 Department of Electronics and Communication Engineering
 Academic Year: 2022 - 2023 (Odd Semester)

INNOVATIVE TEACHING METHOD

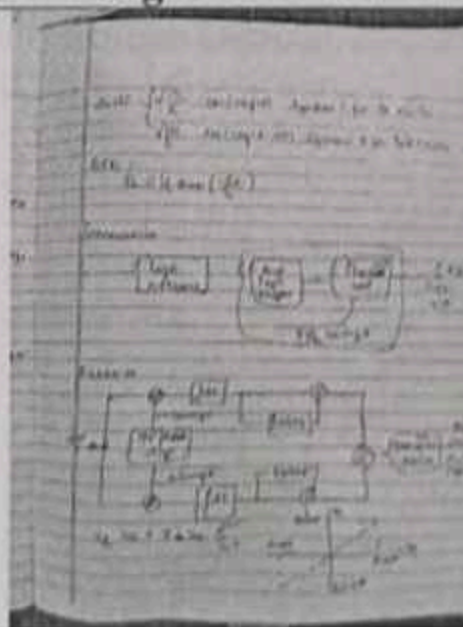
Degree, Semester & Branch: V Semester B.E.ECE A
 Course Code & Title: EC8501 Digital Communication
 Name of the Faculty member: Mr.P.Gunasekaran

Innovative Teaching Method Execution

UNIT V



Hamming Codes - Exit Slip



Prepared by: Mr.P.Gunasekaran
 Date: 06/12/22

Approved by: Dr.B.Deepalakshmi
 Date: 6/12/22

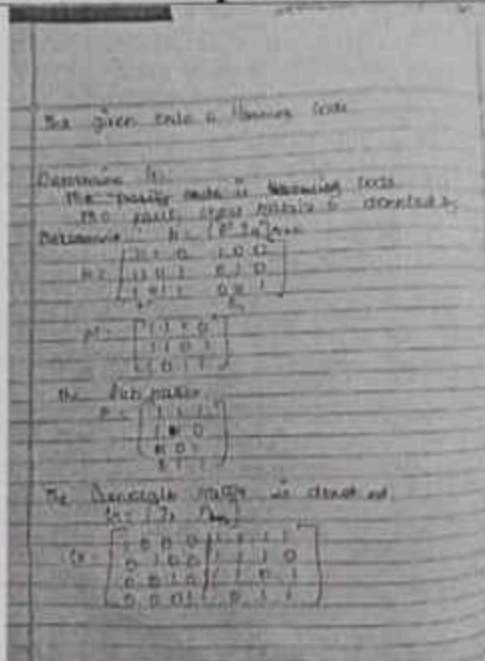
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UNIT IV

BFSK – Open Book Test



BFSK – Open Book Test

See given table a Hamming code

Determine the parity check matrix H and syndrome matrix S for the code

Determine the error correction for the received code

Received code: 1011010

Parity check matrix $H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$

Syndrome matrix $S = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$

The error correction is done as follows:

Received code: 1011010

Parity check matrix $H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$

Syndrome matrix $S = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$

Prepared by: Mr.P.Gunasekaran
 Date: 06/12/22

Approved by: Dr.B.Deeplakshmi
 Date: 6/12/22

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UNIT III

Pulse Shaping – Zero Minute Speech



Pulse Shaping – Zero Minute Speech



Prepared by: Mr.P.Gunasekaran
Date: 06-12-22

Approved by: Dr.B.Deeplakshmi
Date: 6/12/22

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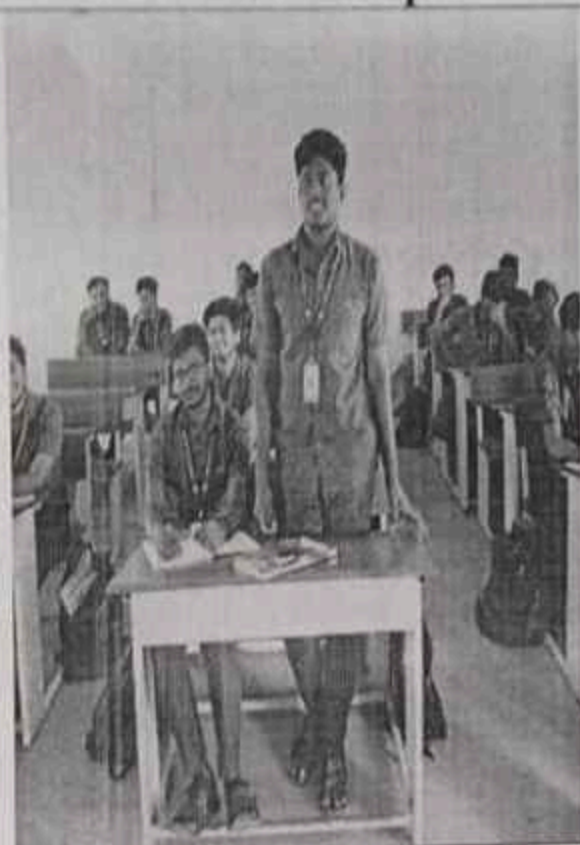
INNOVATIVE TEACHING METHOD

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UNIT II

Bipolar NRZ – JAM on 17.09.2022



Prepared by: Mr.P.Gunasekaran
Date: 27.09.22

Approved by: Dr.B.Deepalakshmi
Date: 27.09.2022

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UNIT I

Information, Entropy – One Minute Paper on 13.08.2022

$\log_a x = \frac{\log x}{\log a}$ $\log_2 10 = \frac{\log 10}{\log 2}$ <u>Entropy</u> → average amount of source information emitted per symbol $H(X) = H = - \sum_{k=0}^{K-1} P_k \log_2 (1/P_k)$ <u>Capacity</u> : intrinsic ability of channel to transmit the information <u>Property :-</u> 1) $I_{sk} = \log_2 \frac{1}{P_k}$ if $P_k \neq 1$ $= \log_2 1$ $= 0$ if $P_k = 1$	<u>Calculation</u> $\log_2 x = \log x / \log 2$ $\log_2 10 = \log 10 / \log 2$ (+) Entropy: Avg amt of information emitted per source (symbol) $H(X) = - \sum_{k=0}^{K-1} P_k \log_2 (1/P_k)$ <u>Capacity</u> : Intrinsic ability of channel to transmit information
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Prepared by: Mr.P.Gunasekaran
 Date: 27.09.22

Approved by: Dr.B.Deepalakshmi
 Date: 27.09.2022